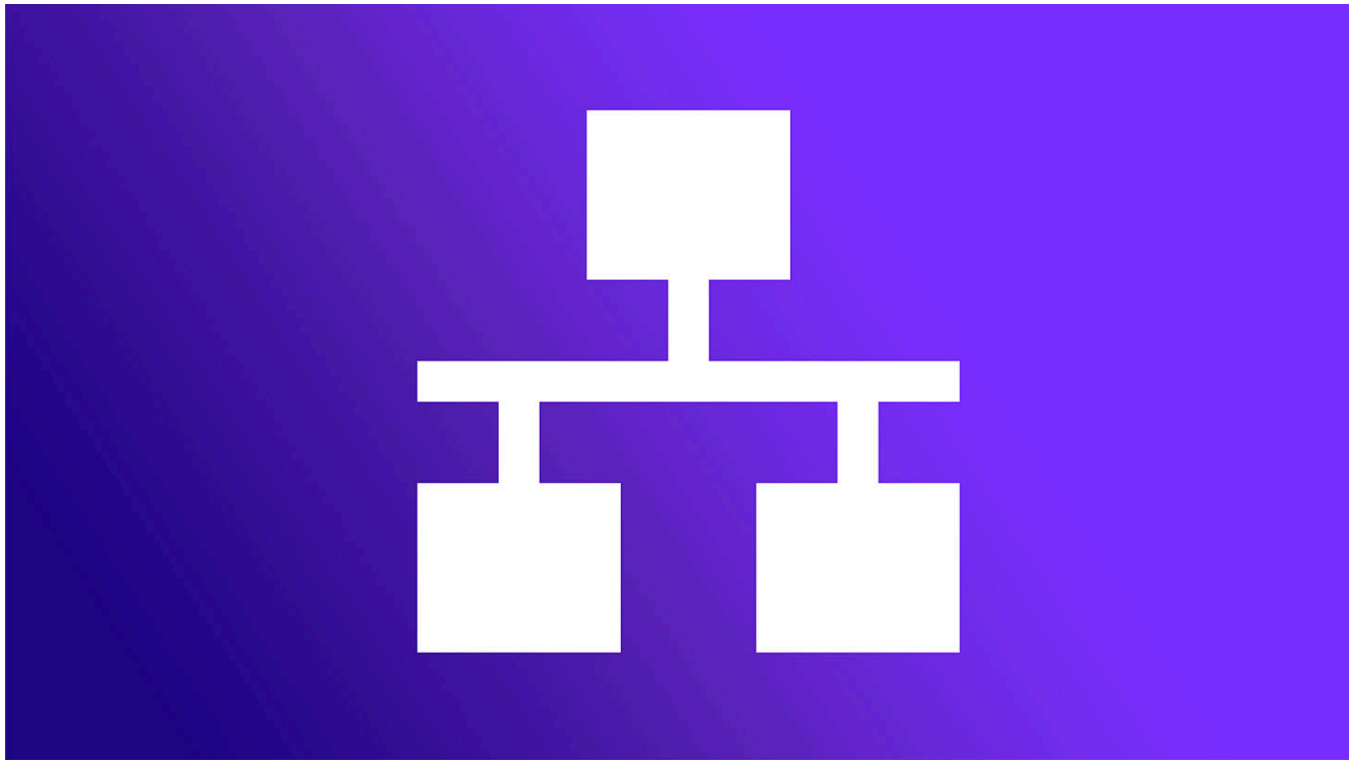
National

Operational Excellence Standard (OES) — Mid-split customer troubleshooting and remediation

HOW22921 | MODIFIED 11/28/2025



Summary

This document defines the actions and tools needed to properly identify and

remediate issues regarding mid-split activation.

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What's changed in this article

Date	Summary of change
November 21, 2025	Made minor updates throughout.

Date	Summary of change
September 16, 2025	Added to Celestial.
April 18, 2024	Conducted annual review and updated. • Updated gateway reference to a Comcast approved DOCSIS 3.1 or greater DOCSIS device. • Removed reference to PHT pass with warnings. • Updated title to OFDMA Partial Bonding. • Removed outdated trouble call flow chart added appendix and reference to related video.
May 18, 2023	Updated mid-split amplifier information.
March 7, 2023	Updated trouble call mitigation flow with mid-split notch filter.
June 1, 2022	Released document.

Overview

This article covers the mid-split customer troubleshooting and remediation OES.

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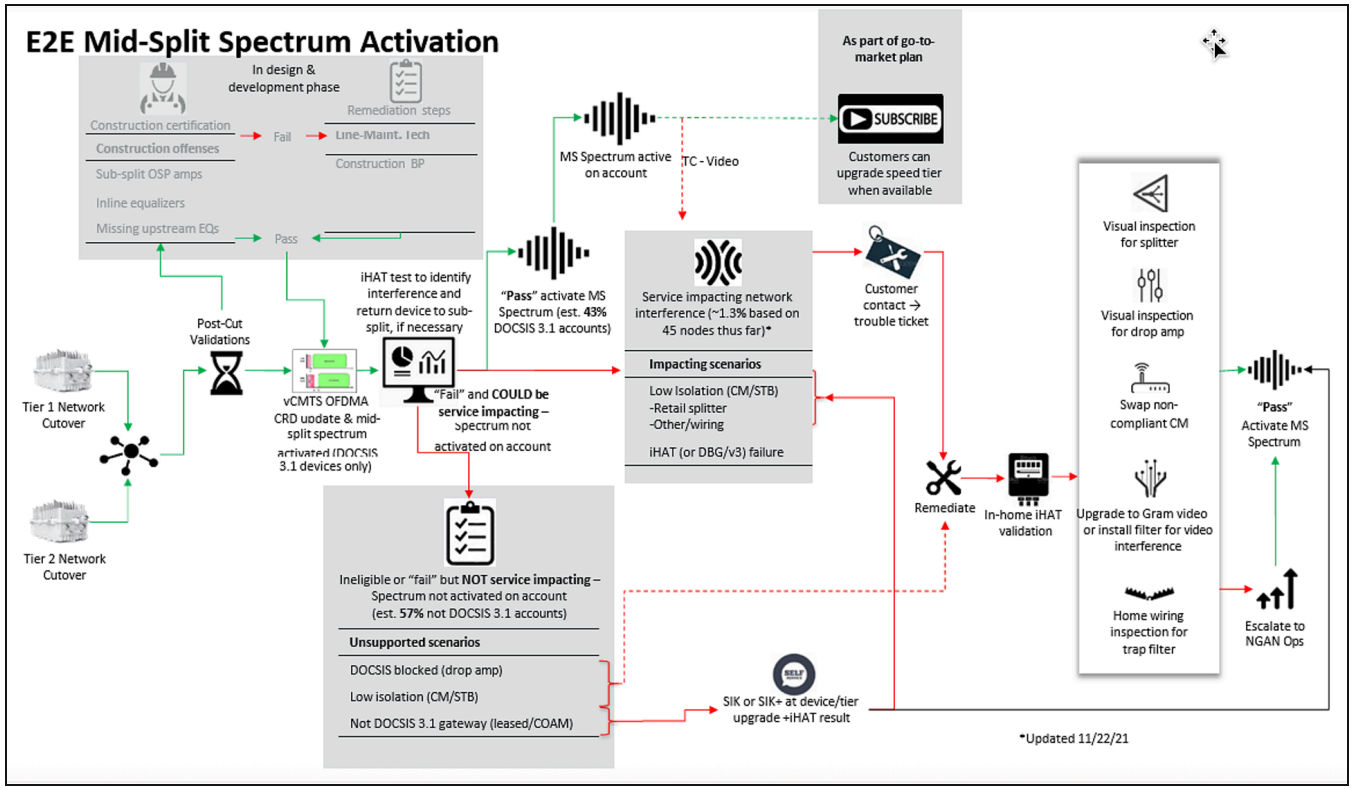
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Required tools and hardware

- ABS mid-split compliant Unity Gain Amplifier — CIFA number 306293

Spectrum activation and home validation

End-to-end mid-split spectrum activation



In-Home Assessment Test (iHAT) tool

iHAT is a CONNECT tool developed to help us determine whether a premise can support mid-split. When a node has been converted to mid-split, iHAT will be run by CONNECT to determine which homes can activate mid-split spectrum without interruption and which homes may need remediation.

iHAT identified issues

Identified issue	Definition	Solution
OFDMA ranging issue or blocked carrier	iHAT determined the mid-split spectrum is blocked an unusable by the gateway	The most likely cause is a drop amplifier in the way and will require a trouble call (TC) to remove or replace with mid-split drop amplifier.
Video interference	iHat identified Video interference from one of the XG devices in the home.	This is caused by low port-to-port isolation in the soft drop system. TC will be required to ensure the passive drop network meets Comcast specifications. (Other solutions available depending on market availability.)
No XG devices	iHAT can perform a test but will not be able to provide video interference information.	
XB3/BWG or older model	The customer does not meet the minimum requirements to perform a test.	

Eligibility requirements for mid-split

- Customers must have a Comcast approved DOCSIS 3.1 or greater DOCSIS device able to tune to the mid-split upstream (OFDMA) spectrum (39.4 –85 MHz).
- To ensure that the home is mid-split compliant, a series of actions and validations must occur to ensure a great customer experience.

Customer TC and mitigation flow: Tiling

1. Verify the customer's complaint.
2. Verify the customer is in an enabled mid-split node.
    - This can be done by looking at the **Plant Information** section of the **Customer Card** in Tech360.
3. Verify the customer has a Comcast-approved DOCSIS 3.1 or greater DOCSIS device.
4. Verify that tiling is present at a lower channel, less than 200 MHz.
5. Power off the mid-split capable gateway and modem.
    - If tilling is still present, then the tiling is not being caused by mid-split. Follow normal troubleshooting procedures to resolve tiling.
    - If there is no tiling, then the tiling is being caused by the mid-split device. Follow the next steps.
6. Follow current **Methods and Procedures** for TC resolution.
7. Verify and correct the customer's drop network to meet OES.
8. Troubleshoot any existing business as usual Premise Health Test (PHT) failures and correct.
    - IF PHT fails for OFDMA, run the OFDMA widget on your XM Meter to locate the failure. If failing at the tap, submit an RTM SA.
9. Once a technician has a **PHT Pass** or **Pass with Overrides** result, the **Mid-split** testing feature will be available.
10. See the remediation options below.
    - **Remediation option 1 — Amplifier present:**
        - Determine the need for an amplifier, remove if possible and replace with:

    - **Passive splitter network** — Follow OES and utilizing OES 1.2 GHz splitter(s).
    - **MoCA passive device** — Follow guidance found in OES — MoCA specifications.
  - **If an amplifier is needed:**
    - Replace the existing sub-split amplifier with a mid-split amplifier.
  - **If this does not resolve the tiling issue:**
    - Proceed to Remediation option 2.
- **Remediation option 1 — Passive network:**
  - Lower loss to common splitter, this will lower transmit power of the XB and raise receive power level of the affected STB/XG.
    - Remove any additional splitters not required.
    - All unused ports should be terminated.
    - Consider utilizing a MoCA passive device with multiple XGs to reduce overall loss when possible. This will lower transmit power of XB and raise the receive power level of the XG device.
  - Increase port-to-port isolation between XB and STB.
    - Remove old splitter(s) and install new OES 1.2 GHz splitter(s).
- **Remediation option 2** — Only required if Remediation option 1 does not eliminate tiling.
  - **QAM X1 customers:**
    - Convert to All IP using available equipment (if enabled in the market).
      - If the customer has a 4K device and wishes to keep 4K availability, they will have to temporarily use an XiD and then call in to have a 4K wireless device shipped (GSK).
      - **Do not** send the customer to a Xfinity store.
  - If the customer is unwilling to give up QAM X1 devices (XG 1 or2), use mid-split notch filter connected directly to the QAM X1 device. This will block any interfering energy from a mid-split enabled gateway.

        - If the customer has multiple QAM X1 devices (XG 1 or2):
            - Based on required tuners, replace any additional QAM X1 devices (XG 1 or2) with XiDs
            - If multiple QAM X1 devices (XG 1 or2) are required to support MoCA outlets, place a mid-split notch filter directly on each QAM X1 device that is affected.
            - Do not connect to MoCA devices like XiD or Xi3.
                - **Do not** connect the notch filter to any XB device.
    - **Legacy and TiVo customers:**
        - Place the mid-split notch filter directly connected to STB. This will block any interfering energy from a mid-split enabled gateway.



11. Limit the use of notch filters.
    - Notch filters are designed for **last resort use only** as they may limit the ability to deliver future technologies.
    - Follow all trouble shooting and remediation steps first in every case. Less than 1% of customers will need a notch filter to mitigate the tiling.
    - Tiling within the bandwidth likely at less than 200 MHz is the only condition that a notch filter will correct. Tiling in other bands, slow speeds, and latency are not conditions that will be corrected by a notch filter.
    - Use Resolution code **INST MIDSPLIT FILTER** when installing a notch filter.

Customer TC and mitigation flow: OFDMA partial bonding (formally blocked carrier)

1. Verify the customer's issue.
2. Verify the customer is in an enabled mid-split node.
    - This can be done by looking at the **Plant Information** section of the **Customer Card**.
3. Verify the customer has a Comcast-approved DOCSIS 3.1 or greater DOCSIS device able to tune to the Mid-Split upstream (OFDMA) spectrum (39.4–85 MHz).
4. Follow current **Methods and Procedures** for TC resolution.
5. Verify and correct the customer's drop network to meet OES.
6. Troubleshoot any existing business as usual PHT failures and correct.

The steps below define the actions needed to properly mitigate the customer's network and they should be applied in order of preference.

- **Option 1:** Remove the drop amplifier and reconfigure the customers network using only passive splitters.
- **Option 2:** Remove the drop amplifier and reconfigure the customers network using a MoCA passive splitter following all guidance from OES — MoCA specifications.
- **Option 3:** Replace the existing sub-split amplifier with a mid-split amplifier.

Drop amp matrix

	Unity gain port limitations	Passive port limitations
Passive ports only (Gen 1)	Will **not** pass mid-split upstream (OFDMA) signal to the unity gain ports	Will not pass MoCA signals between unity gain ports and passive port
Partial MoCA (Gen 1.5)		Will pass MoCA signals between select unity gain ports and passive port
Full MoCA (Gen 2)		Will pass MoCA signals between unity gain ports and passive port
Mid-split amps	None	None

Mid-split amplifier



Appendix

Please see the video below.

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